

CivilQntify User Manual

Complete Guide to Using the CivilQntify Desktop Application Version 1.0.10

Concrete Mix Design, Particle Size Distribution, Material Quantification, Cost Estimation, and History

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1 Introduction

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This manual explains how to use every feature of the CivilQntify desktop application through its graphical user interface. It is written for civil engineers, concrete technologists, quantity surveyors, and students who proportion concrete mixes, check aggregate grading, quantify structural materials, estimate project cost, and review saved work.

CivilQntify automates mix proportioning to three design codes, ACI 211.1-22, IS 10262:2019, and BRE 331:1997 DOE method, analyses aggregate particle size distribution against IS 383:2016 or ASTM C33/C33M grading bands, computes material bills for structural elements or work volumes, estimates cost in Ghana cedis (GHc), and keeps a searchable history of every calculation with PDF, CSV, JSON, and text exports.

The fastest way to work is Mix Design to Material Quantification to Cost Estimation, using the Send buttons that carry results forward. Each tab also works standalone when quantities or prices are entered manually. Figures in this manual are screenshots of the running application.

2 Getting Started

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2.1 System Requirements and Installation

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- Operating system: Windows 10 or later, Ubuntu 22.04 or later, or macOS 13 or later.
- Python 3.10 or later with the project requirements installed, or the packaged CivilQntify installer.
- Launch the graphical interface with `python main.py` from the project folder, or use `python main_cli.py` for the interactive command line workflow.

2.2 Main Window Overview

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The main window has a top toolbar, a tab bar with four tabs, and a status bar. The toolbar holds the Weather button and the settings gear. The tabs are Mix Design, Material Quantification, Cost

Estimation, and History. The status bar at the bottom confirms actions such as transfers and exports.

- Mix Design: proportion concrete mixes and analyse aggregate grading on two inner tabs, PSD and Concrete Mix Design.
- Material Quantification: convert a mix into material bills by volume, elements, or nominal mix ratios.
- Cost Estimation: price the bill with manual material rates and overheads.
- History: find, reload, and export any past calculation.

The screenshot displays the 'Mix Design' software interface. The top navigation bar includes 'Weather' and a settings gear icon. The main window is divided into four tabs: 'Mix Design' (active), 'Material Quantification', 'Cost Estimation', and 'History'. Within the 'Mix Design' tab, there are two sub-tabs: 'PSD' and 'Concrete Mix Design' (active). The 'Concrete Mix Design' sub-tab is further divided into 'STEP 1 — DESIGN PARAMETERS' and 'MATERIAL QUANTITIES (PER M³)'. The 'DESIGN PARAMETERS' section includes dropdowns for 'STANDARD' (IS 10262:2019), 'CALCULATION' (Concrete Mix Design), 'CONCRETE TYPE' (Reinforced Concrete), 'EXPOSURE CLASS (IS 456)' (None), 'PLACING METHOD' (Chute (non-pumpable)), and 'SITE CONTROL' (Good site control). It also features a 'MAX FREE W/C RATIO' button, a 'MASS CONCRETE' checkbox, and a warning box about grade and exposure. At the bottom of this section are 'Calculate Mix Design' and 'Clear' buttons. The 'MATERIAL QUANTITIES' section shows input fields for 'CEMENT', 'WATER', 'FINE AGGREGATE', 'COARSE AGGREGATE', 'SCM', 'ADMIXTURE', 'W/C RATIO', and 'AIR CONTENT', along with a 'TARGET STRENGTH' field. Below this is a 'CALCULATION STEPS' table with columns for '#', 'DESCRIPTION', 'FORMULA', 'RESULT', 'UNIT', and 'REFERENCE'. At the bottom of the interface, there are 'Export CSV', 'Export PDF', and 'Send to Quantification' buttons. A status bar at the very bottom shows: 'Quantified — Gross: 10.500 m³ | Cement: 3,675.0 kg (74 bags) | Water: 1,837.5 kg'.

Figure 1. Main window on the Mix Design tab with the Concrete Mix Design inner tab open.

2.3 Toolbar Settings and Weather

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Select the gear icon at the top right to open Settings. The dialog controls the unit system, Metric or Imperial, and the visibility of the Weather button. Unit choices relabel every quantity spin box across the application. Select the Weather button to open the weather panel, which shows current temperature and humidity for the selected Ghana city with concreting advice for hot or cold weather placing.

- Open Settings with the gear icon, change units or weather visibility, then select Apply.
- Open weather with the Weather button, pick a city, and read the placing advice before scheduling a pour.

3 Concrete Mix Design Tab

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Open the Mix Design tab and choose the Concrete Mix Design inner tab. The left panel is a scrollable form in three steps plus optional checks. The right panel shows material quantity cards, the target strength card, the step by step calculation table with formula, result, unit, and standard reference columns, and the action buttons Export CSV, Export PDF, and Send to Quantification.

3.1 Step 1 Design Parameters

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1. Standard: select IS 10262:2019, ACI 211.1-22, or DOE BRE 331:1997. Exposure, water content, and aggregate tables follow the selected code.
2. Calculation: select Concrete Mix Design for normal proportioning or Strength Prediction to reverse calculate 28 day strength from a water cement ratio.
3. Concrete Type: select Plain, Reinforced, or Prestressed. Durability limits follow the choice.
4. Exposure Class (IS 456): select the exposure condition, for example Mild, Moderate, Severe, or None. Minimum grade and maximum free water cement ratio are gated by this selection.
5. Max Free W/C Ratio: read only display of the durability cap for the chosen exposure.
6. Placing Method: select Chute non-pumpable or Pumpable, which adjusts water and aggregate inputs.
7. Site Control: select Good or Poor site control, which sets the assumed standard deviation used in target strength.
8. Mass Concrete: tick for mass pours, which applies Annex F adjustments.
9. Characteristic Strength f_{ck} (MPa): enter the specified 28 day strength, 5 to 100 MPa.
10. Slump (mm): enter the required workability.
11. Select Calculate Mix Design to run the design, or Clear to reset the form.

3.2 Step 2 Material Properties

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Enter cement type and grade, specific gravities of cement, fine aggregate, and coarse aggregate, water absorption and free moisture of each aggregate, nominal maximum size of coarse aggregate, fineness modulus or zone, and air content where applicable. Moisture entries correct batch water automatically. Select an information icon beside any field for the standard clause behind it.

3.3 Step 3 Admixtures and Supplementary Cementitious Materials

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Enter superplasticizer dosage with its water reduction percentage (IS 10262 Annex G), and fly ash or GGBS replacement percentage with its efficiency factor. The engine recomputes cementitious content, water content, and paste volume, and the calculation steps table cites each adjustment.

3.4 Optional Trial and Self-Compacting Checks

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The Trial Check group (ACI Section 5.3.10) accepts a trial water cement ratio to compare against the design value. The SCC fresh property group (IS) checks slump flow, V-funnel, and L-box criteria. Trial mix workbooks for DOE Section 6 and ACI Appendix A.5 open from the result panel after a design, with batch schedule, feedback inputs, density correction, and the verdict of minor adjustment or second trial.

3.5 Results and Exports

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- Material cards: read cement, water, fine aggregate, coarse aggregate, SCM, admixture, water cement ratio, air content, and target strength per cubic metre.
- Calculation Steps table: audit every step with its formula and standard reference.
- Export CSV: save the quantities and steps for site logs. Export PDF: preview the formatted report, then save it with embedded inputs, steps, and charts.
- Send to Quantification: transfer the per cubic metre mix into the Material Quantification tab and continue to Section 5.

4 Particle Size Distribution PSD Tab

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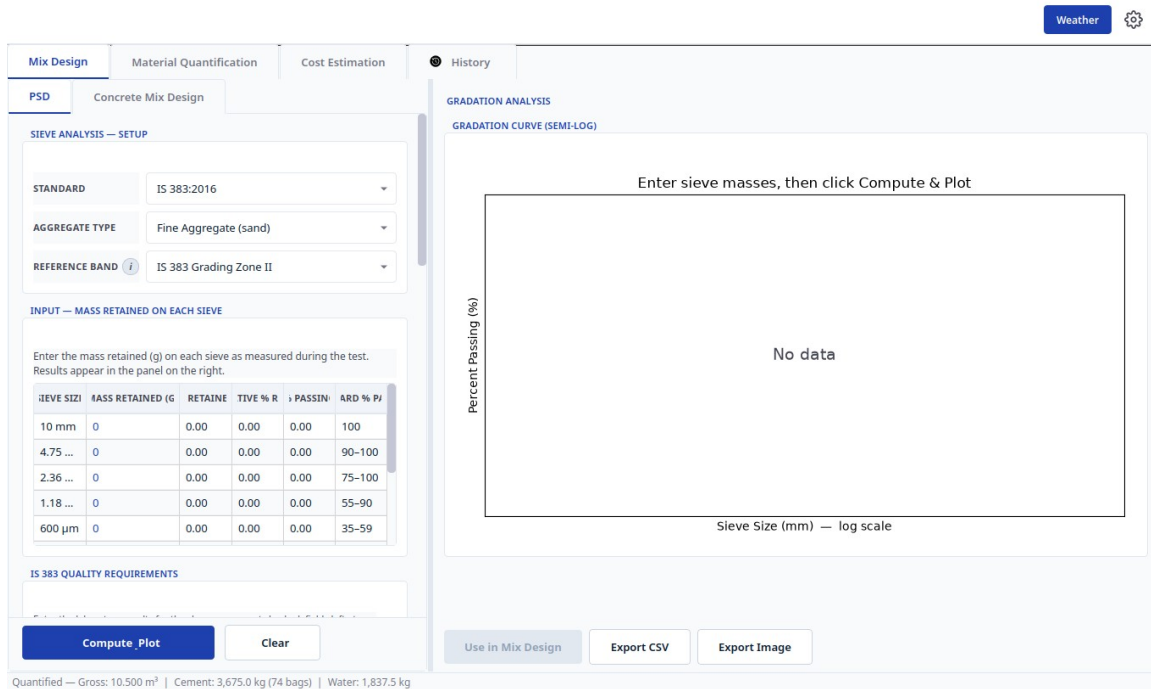


Figure 2. PSD inner tab with sieve setup on the left and the semi-log gradation curve on the right.

Open the PSD inner tab inside Mix Design before designing a mix when sieve data is available, because derived values can feed the mix form. The left panel holds Sieve Analysis Setup, the retained mass input table, and quality requirements. The right panel shows the gradation curve, the results summary, and the action buttons Use in Mix Design, Export CSV, and Export Image.

1. Standard: select IS 383:2016 or ASTM C33/C33M. Never mix sieve series and limits across standards; rows and bands always follow this selector.
2. Aggregate Type: select Fine Aggregate (sand) or Coarse Aggregate.
3. Reference Band: select the grading reference, for example IS 383 Grading Zone II for sand, or the 10 mm, 20 mm, or 40 mm coarse bands which map to ASTM Sizes 8, 67, and 467.
4. Input table: enter the mass retained in grams on each sieve exactly as weighed. IS fine work uses a 10 mm top sieve with Zones I to IV; ASTM fine work uses a 9.5 mm top sieve with the Table 1 envelope; coarse tables list the full Table 7 or Table 2 sieve series.
5. Select Compute and Plot. The app computes percent retained, cumulative passing, fineness modulus, D10, D30, D60, and the zone or envelope verdict, then draws the measured curve over the smoothed standard band on a semi-log plot.
6. Read compliance in the results summary. For ASTM work, select the compliance control to open the full requirement dialog covering grading, deleterious substances, and soundness with exact clause citations.
7. Select Use in Mix Design to lock derived fineness modulus, zone, 600 micron passing, or nominal size into the mix form. Select Clear on the PSD tab to release the lock and restore previous values.

8. Export CSV saves the sieve table and results; Export Image saves the gradation chart.

4.2 Passing Test Results

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The four worked examples below were entered into the running application as laboratory style retained masses, one for fine and one for coarse aggregate under each standard. Every one conforms to its selected band: the results summary shows a green Conforms to selected band banner, the measured curve stays inside the shaded standard band, D10, D30, and D60 markers are drawn on the curve, and the status bar confirms that all applicable requirements were checked with no non-conformance found.

Case	Standard and Reference Band	Sample Mass	Application Result
Fine sand, IS 383 Zone II	IS 383:2016, Grading Zone II	1000 g total	Fineness modulus 2.74, Zone II, conforms
Fine sand, ASTM Table 1	ASTM C33/C33M, Fine Table 1	1000 g total	Fineness modulus 2.80, conforms
Crushed stone, IS graded 20 mm	IS 383:2016 Table 7, graded 20 mm	5000 g total	D60 12.5 mm, conforms
Crushed stone, ASTM Size 67	ASTM C33/C33M Table 2, Size 67 (20 mm)	5000 g total	D60 11.5 mm, conforms

Retained masses used, coarsest sieve first with the pan catch last. IS 383 fine Zone II (10, 4.75, 2.36, 1.18, 600 micron, 300 micron, 150 micron): 0, 50, 70, 160, 250, 280, 140 g with a 50 g pan. ASTM C33 fine Table 1 (9.5, 4.75, 2.36, 1.18, 600 micron, 300 micron, 150 micron): 0, 20, 80, 220, 260, 250, 120 g with a 50 g pan. IS 383 coarse graded 20 mm (80 down to 2.36 mm): 0, 0, 0, 250, 750, 1000, 1000, 1750, 200 g with a 50 g pan. ASTM C33 coarse Size 67 (100 mm down to 300 micron): 0, 0, 0, 0, 0, 0, 0, 0, 250, 1250, 1600, 1650, 150, 50, 0 g with a 50 g pan.

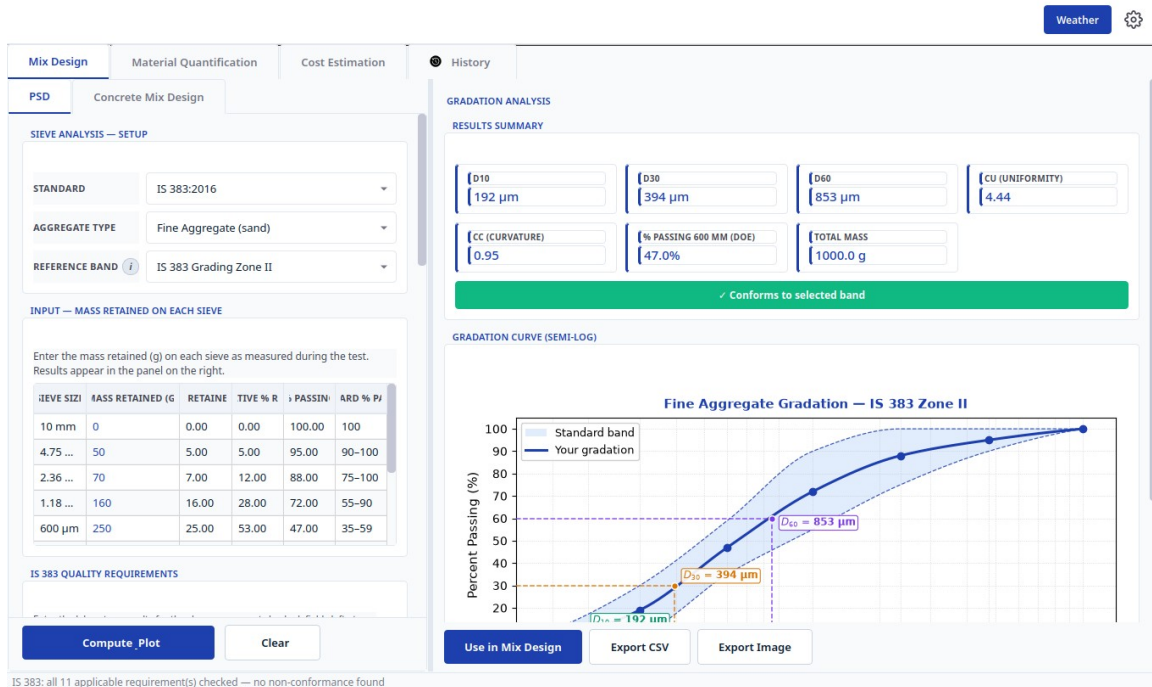


Figure 3. Passing result for IS 383 fine aggregate, Zone II. Green banner, curve inside the band, and D10, D30, D60 markers.

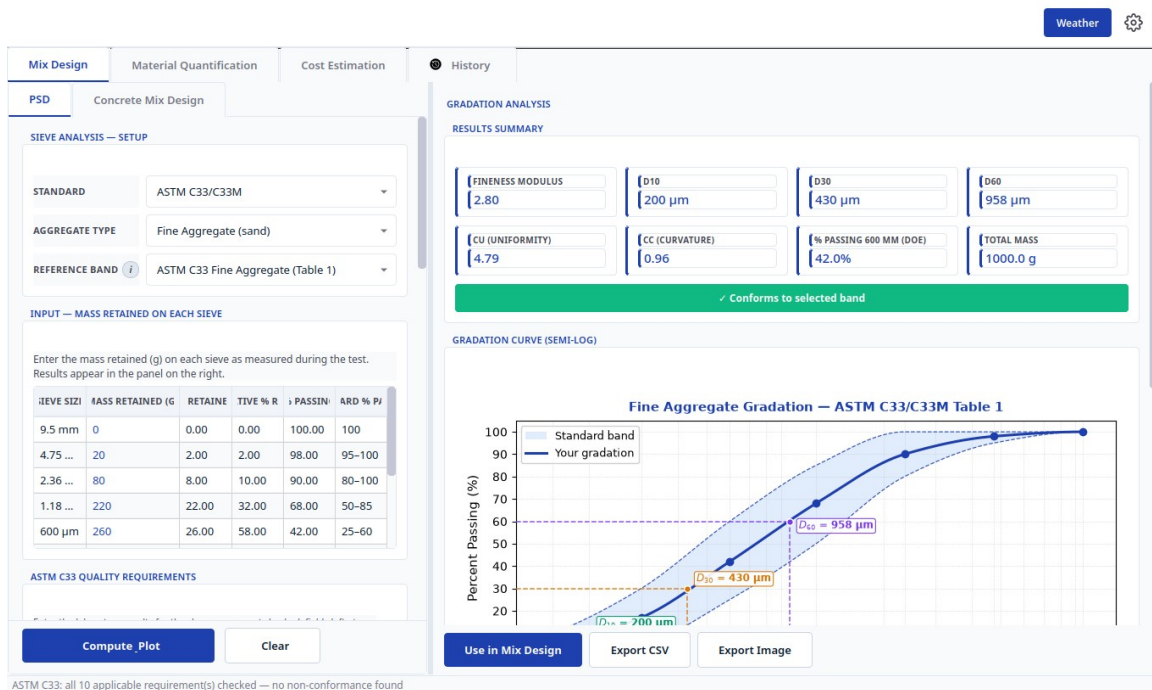


Figure 4. Passing result for ASTM C33 fine aggregate, Table 1, with fineness modulus 2.80.

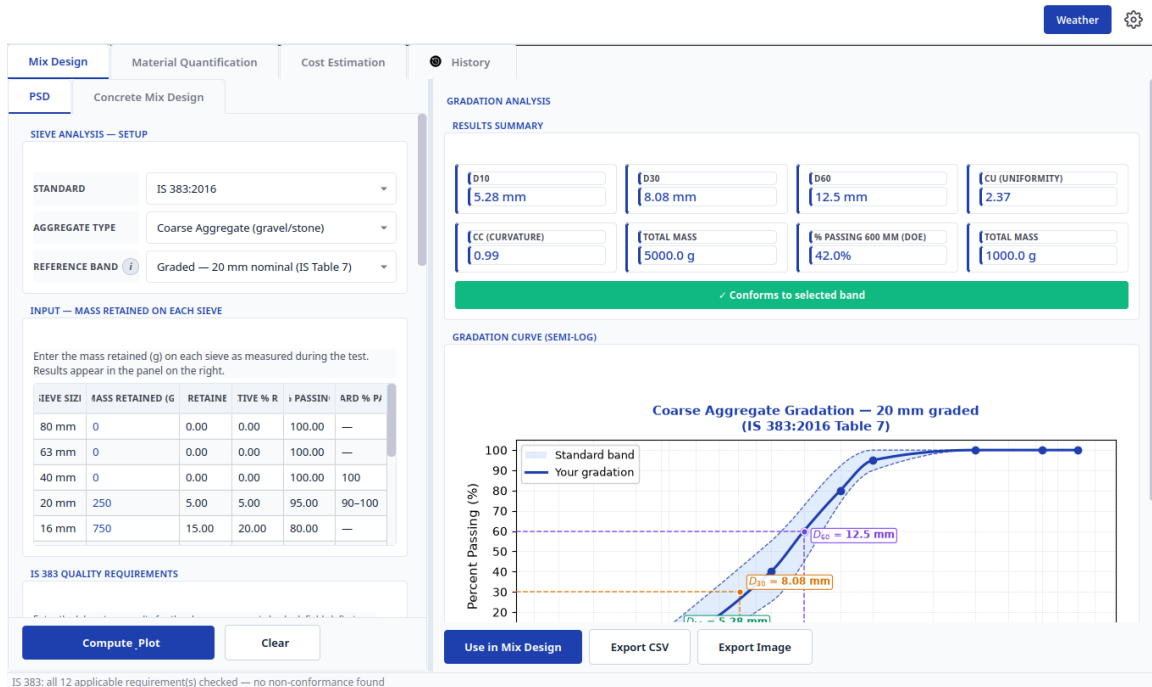


Figure 5. Passing result for IS 383 coarse aggregate, graded 20 mm nominal, Table 7.

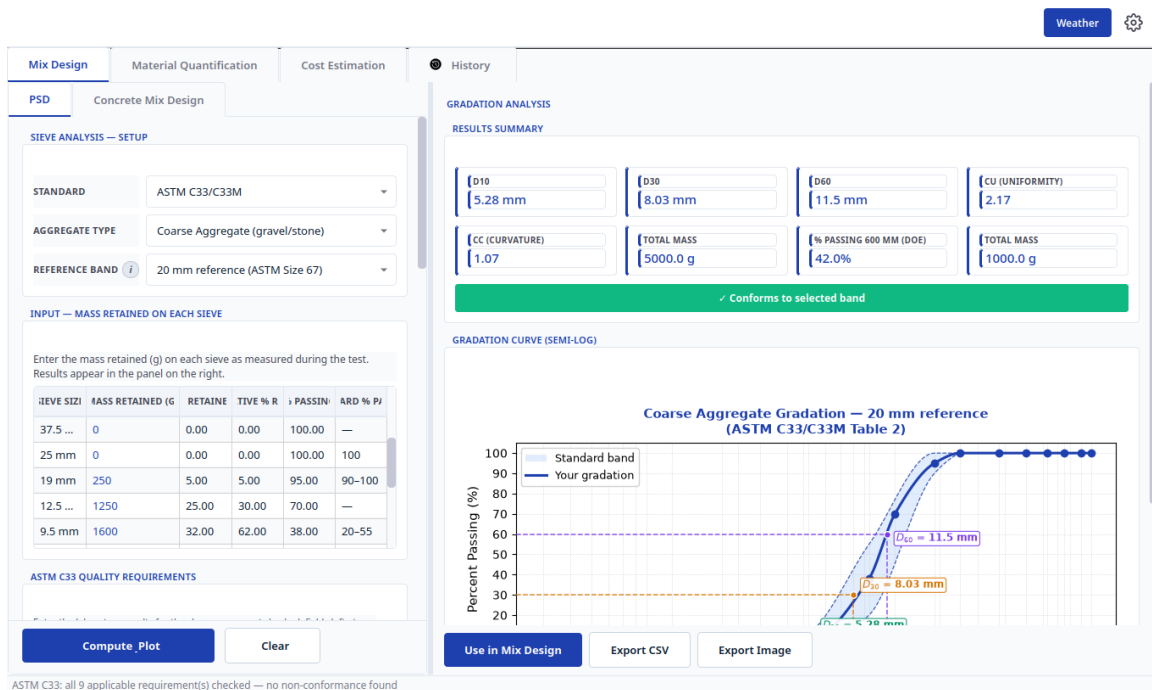


Figure 6. Passing result for ASTM C33 coarse aggregate, 20 mm reference Size 67, Table 2.

To repeat any passing test, select the standard, aggregate type, and reference band exactly as shown in the figure, type the retained masses into the input table, and select Compute and Plot.

Select Use in Mix Design to lock the derived values into the mix form, Export CSV for the sieve record, or Export Image for the chart.

4.3 Failing Test Results and What They Mean

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A non-conforming sample is unmistakable. The results summary shows a red banner counting the sieves out of band, the offending points on the curve are marked with red crosses outside the shaded band, and a compliance dialog opens automatically, citing the exact clause and the measured value against the limit. Two failing examples, one per standard, are shown below.

IS 383 Fine Aggregate Failure, Too Coarse

Retained masses 0, 150, 250, 300, 200, 80, 15 g with a 5 g pan (1000 g total) give passing values of 85, 60, 30, 10, 2.0, and 0.5 percent against Zone II minima of 90, 75, 55, 35, 8, and 0 percent. Five sieves fall out of band, the fineness modulus rises to 4.12, and the dialog reports that Clause 6.3 Table 9 is not met, naming each sieve with its measured passing value.

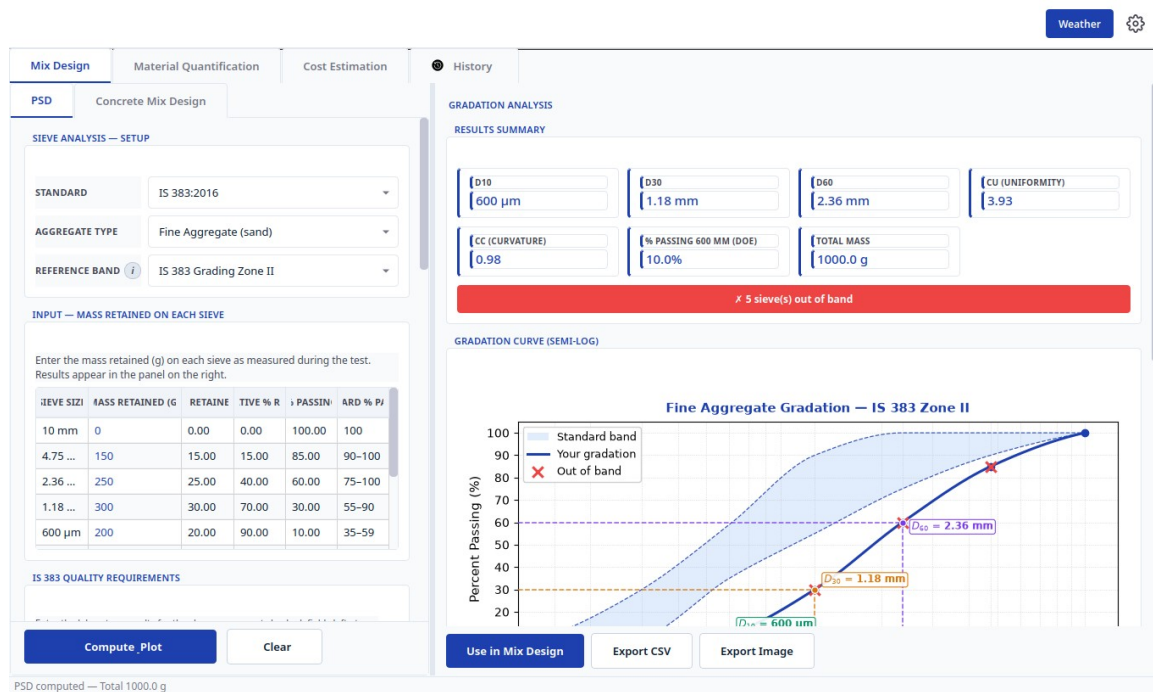


Figure 7. Failing IS 383 Zone II sand. Red banner counts five sieves out of band and red crosses mark the breaches.

⚠ 1 requirement(s) NOT met — the fine aggregate analysis does not conform to IS 383.

The analysis does not meet the following requirements:

Clause 6.3, Table 9 — Grading requirements (sieve analysis): 3 sieve(s) out of zone.

4.75 mm 85.0 % passing is below the Zone I limit of 90 %; 600 µm 10.0 % passing is below the Zone I limit of 15 %; 300 µm 2.0 % passing is below the Zone I limit of 5 %.

STATUS	CLAUSE	REQUIREMENT	RESULT / MEASURED VALUE
✗ NOT ...	6.3, Table 9	The grading of fine aggregate shall ...	3 sieve(s) out of zone
— n/a	Table 2 SI (i), Clause 5.2.1	Not more than 1 % by mass (IS 2386 ...	not tested
— n/a	Table 2 SI (ii), Clause 5.2.1	Not more than 1 % by mass (IS 2386 ...	not tested
— n/a	Table 2 SI (iii), Clause 5.2.1	Not more than 3 % by mass (IS 2386 ...	not tested
— n/a	Table 2 SI (v), Clause 5.2.1	Not more than 1 % by mass (IS 2386 ...	not tested
— n/a	Table 2 SI (vi), Clause 5.2.1	Total of all deleterious materials ...	not tested
— n/a	Table 2 Note 3, Clause 5.2.1	Mica limited to 1.00 % by mass wher...	not tested
— n/a	5.2 Note 4	Aggregate shall not contain harmful...	not tested
— n/a	5.3	Combined flakiness and elongation ...	not tested
— n/a	5.5.1	For concrete liable to frost action th...	not tested
— n/a	5.6	Aggregate shall comply with the IS ...	not tested

Clause and table numbers follow IS 383:2016 (Third Revision, incorporating Amendment No. 1), extracted in docs/IS-383-2016-Coarse-and-Fine-Aggregate-for-Concrete.md. Requirements shown as "n/a" had no test result entered or carry no requirement (a dash cell of the standard's tables) and were not evaluated.

✓ OK

Figure 8. IS 383 compliance dialog. Clause 6.3 Table 9 is cited with measured passing values per sieve.

ASTM C33 Coarse Aggregate Failure, Too Fine

Retained masses of 100 g on 19 mm, 300 g on 12.5 mm, 900 g on 9.5 mm, 1500 g on 4.75 mm, 1200 g on 2.36 mm, 600 g on 1.18 mm, and 200 g on 300 micron with a 200 g pan (5000 g total) give 74 percent passing 9.5 mm against a 20 to 55 percent limit, 44 percent passing 4.75 mm against a 0 to 10 percent limit, and 20 percent passing 2.36 mm against a 0 to 5 percent limit. Three sieves fall out of band and the dialog cites Clause 10.1 Table 2 with the measured values.

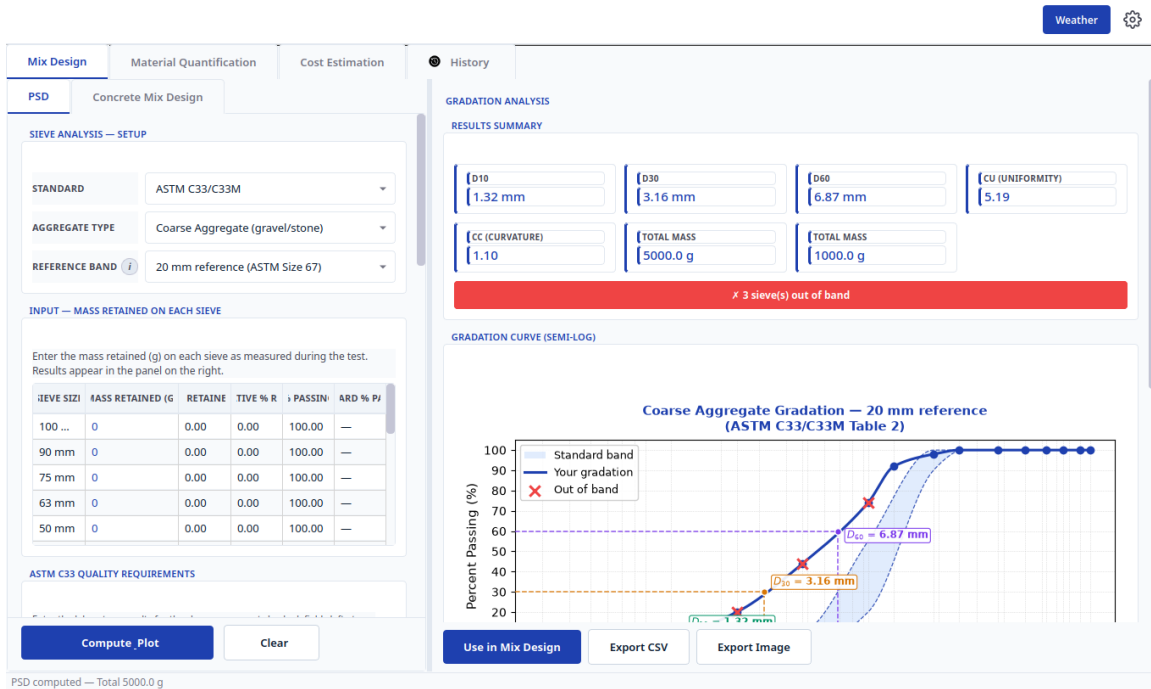


Figure 9. Failing ASTM Size 67 stone. The excess fines pull the curve left of the band with three sieves out.

⚠ 1 requirement(s) NOT met — the coarse aggregate analysis does not conform to ASTM C33.

The analysis does not meet the following requirements:

Clause 10.1 (Table 2) — Grading requirements (size number): 3 sieve(s) out of band.

9.5 mm: 74.0 % passing vs 20–55 % limit (too fine, max 55 %); 4.75 mm: 44.0 % passing vs 0–10 % limit (too fine, max 10 %); 2.36 mm: 20.0 % passing vs 0–5 % limit (too fine, max 5 %).

STATUS	CLAUSE	REQUIREMENT	RESULT / MEASURED VALUE
✗ NOT ...	10.1 (Table 2)	Grading shall be within the limits of t...	3 sieve(s) out of band
— n/a	Table 3 (11.1)	Class 3S: not more than 5 % clay lum...	not tested
— n/a	Table 3 (11.1)	Class 3S: not more than 5 % chert	not tested
— n/a	Table 3 (11.1)	Class 3S: not more than 7 % sum of ...	not tested
— n/a	Table 3 (11.1), Footnote C	Class 3S base limit 1.0 %	not tested
— n/a	Table 3 (11.1)	Class 3S: not more than 0.5 % coal an...	not tested
— n/a	Table 3 (11.1), Footnote A	Abrasion loss not greater than 50 % ...	not tested
— n/a	Table 3 (11.1), Footnote B	Weighted average loss not greater ...	not tested
— n/a	11.2	Aggregate for concrete subject to ...	not tested

Clause and table numbers follow ASTM C 33 – 99ae1, the edition extracted in docs/ASTM-C33-99-Concrete-Aggregates.md (later C33/C33M editions renumber the Clause 6.1 grading table as “Table 1”). Requirements shown as “n/a” had no test result entered and were not evaluated.

OK

Figure 10. ASTM C33 compliance dialog. Clause 10.1 Table 2 is cited with measured values and the too fine verdict.

When a result fails, select OK to close the dialog and read the red banner and curve markers. Do not transfer failing values with Use in Mix Design until the material position is resolved. The standard site responses are to retest a fresh sample, blend with a complementary stockpile and re-run the analysis, or reject the delivery. Rows marked n/a in the dialog had no test result entered and were not evaluated.

5 Material Quantification Tab

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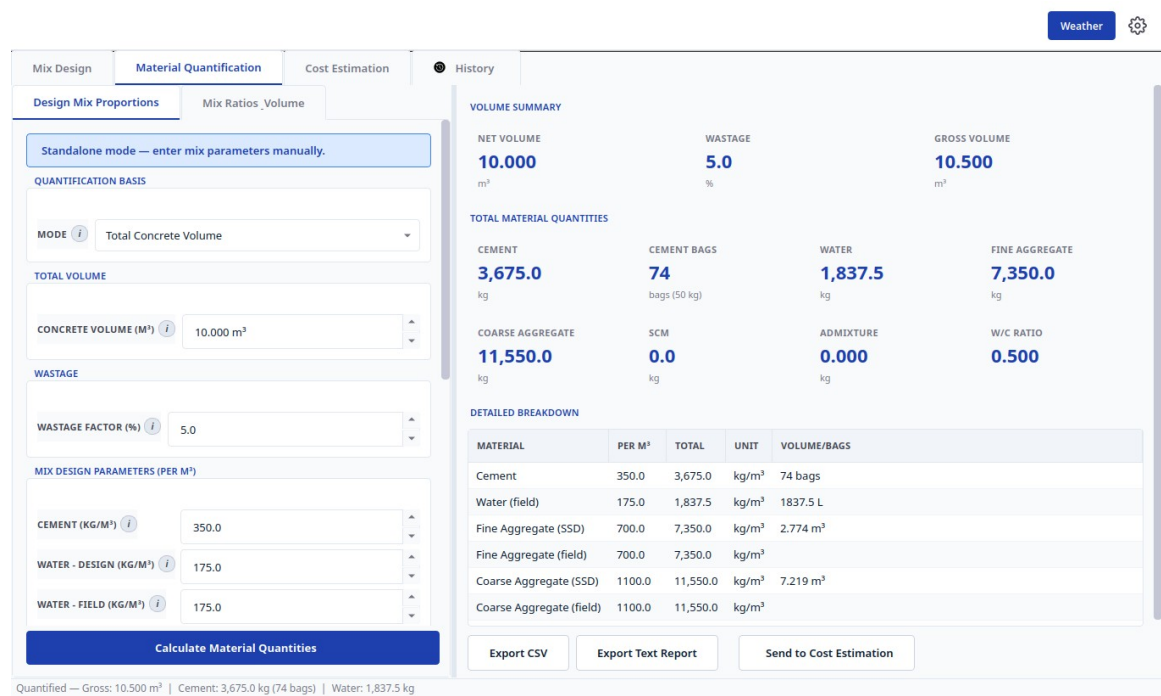


Figure 11. Material Quantification on Design Mix Proportions with volume summary and detailed breakdown.

Open the Material Quantification tab to turn a mix into a site bill. It has two inner tabs. Design Mix Proportions uses an engineered per cubic metre mix, either transferred from Mix Design or typed manually. Mix Ratios and Volume uses a nominal Cement to Sand to Coarse Aggregate ratio with a work volume.

5.1 Design Mix Proportions Mode

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1. Quantification Basis, Mode: select Total Concrete Volume or Structural Elements. In Elements mode, add slabs, beams, columns, footings, walls, retaining walls, and staircases with dimensions; the app totals the net volume.
2. Total Volume: enter the concrete volume in cubic metres.
3. Wastage: enter the wastage factor percent, typically 5 percent; gross volume equals net volume plus wastage.
4. Mix Design Parameters (per cubic metre): confirm or type cement, design water, field water, fine and coarse aggregate, SCM, and admixture. Transferred mixes arrive filled; standalone work starts from the defaults.
5. Select Calculate Material Quantities. Read Net Volume, Wastage, and Gross Volume, then the total cards for cement, cement bags, water, fine and coarse aggregate, SCM, admixture, and water cement ratio, then the Detailed Breakdown table with per cubic metre, total, unit, and volume or bag columns.
6. Export CSV or Export Text Report for the site, or select Send to Cost Estimation to carry the bill forward.

5.2 Mix Ratios and Volume Mode

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The screenshot displays the 'Mix Ratios_Volume' tab in a software application. The interface is divided into several sections:

- Design Mix Proportions:** Includes a note about Mix Ratio Mode and a 'VOLUME OF WORK BASIS' section with a dropdown for 'MODE' set to 'Total Work Volume'.
- TOTAL VOLUME OF WORK:** A field for 'WORK VOLUME (M³)' set to '10.000 m³'.
- MIX PROPORTIONS (CEMENT : SAND : COARSE AGG):** A section with input fields for 'STANDARD PRESET' (M20 (1:1.5:3)), 'CEMENT RATIO (PARTS)' (1.00), 'FINE AGG / SAND (PARTS)' (1.50), 'COARSE AGG / STONE (PARTS)' (3.00), and 'WATER-CEMENT RATIO (W/C)' (0.500). A 'Calculating...' button is at the bottom.
- VOLUME SUMMARY:** A summary of key values:

NET VOLUME	WASTAGE	GROSS VOLUME
10.000 m³	5.0 %	10.500 m³
- TOTAL MATERIAL QUANTITIES:** A grid of material quantities:

CEMENT	CEMENT BAGS	WATER	FINE AGGREGATE
3,675.0 kg	74 bags (50 kg)	1,837.5 kg	7,350.0 kg
COARSE AGGREGATE	SCM	ADMIXTURE	W/C RATIO
11,550.0 kg	0.0 kg	0.000 kg	0.500
- DETAILED BREAKDOWN:** A table showing material breakdown:

MATERIAL	PER M³	TOTAL	UNIT	VOLUME/BAGS
Cement	350.0	3,675.0	kg/m³	74 bags
Water (field)	175.0	1,837.5	kg/m³	1837.5 L
Fine Aggregate (SSD)	700.0	7,350.0	kg/m³	2.774 m³
Fine Aggregate (field)	700.0	7,350.0	kg/m³	
Coarse Aggregate (SSD)	1100.0	11,550.0	kg/m³	7.219 m³
Coarse Aggregate (field)	1100.0	11,550.0	kg/m³	
- Buttons:** 'Export CSV', 'Export Text Report', and 'Send to Cost Estimation'.
- Footer:** 'Quantified — Gross: 10.500 m³ | Cement: 3,675.0 kg (74 bags) | Water: 1,837.5 kg'.

Figure 12. Mix Ratios and Volume mode with a nominal 1 to 1.5 to 3 preset and work volume basis.

Select the Mix Ratios and Volume inner tab, choose Total Work Volume or Elements for the volume basis, enter the work volume, then set Cement to Sand to Coarse Aggregate parts directly

or from a standard preset such as M20 (1:1:1.5:3). Enter the water cement ratio and factors such as dry volume factor and bag volume (1 bag equals 0.035 cubic metres). Select Calculate, then export or send to Cost Estimation exactly as in Section 5.1.

6 Cost Estimation Tab

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Weather

Mix DesignMaterial QuantificationCost EstimationHistory

Standalone mode — enter material quantities manually.

MATERIAL QUANTITIES

CEMENT (BAGS)

0

FINE AGG (M³)

0.000 m³

COARSE AGG (M³)

0.000 m³

WATER (L)

0.0 L

SCMS (KG)

0.0 kg

ADMIX (KG)

0.000 kg

TOTAL VOLUME

1.00 m³

WASTAGE

5.0%

MATERIAL PRICES (GH¢)

CEMENT (PER BAG)

GH¢ 85.00

Estimate Project Cost

COST SUMMARY

MATERIAL COST / M³

TOTAL MATERIAL COST

TOTAL PROJECT COST

COST / BAG OF CONCRETE

MATERIAL COST BREAKDOWN

MATERIAL	QTY	UNIT	UNIT PRICE (GH¢)	TOTAL COST (GH¢)
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PROJECT SUMMARY

Export Cost Report (CSV)

Export Cost Report (PDF)

Quantified — Gross: 10.500 m³ | Cement: 3,675.0 kg (74 bags) | Water: 1,837.5 kg

Figure 13. Cost Estimation with material quantities, prices in Ghana cedis, and summary cards.

Open the Cost Estimation tab to price the bill. A banner states whether quantities arrived from Quantification or are being entered manually in standalone mode. The left panel holds Material Quantities, Material Prices (GHc), Additional Costs, and Project Information. The right panel shows the Cost Summary cards, the Material Cost Breakdown table, the Project Summary, and the export buttons.

1. Material Quantities: confirm transferred cement bags, fine and coarse aggregate volumes, water, SCMs, admixture, total volume, and wastage, or type them manually in standalone mode.
2. Material Prices (GHc): enter cement per bag, sand and gravel per cubic metre, water, and admixture rates.
3. Additional Costs: enter labour, transport, overheads, profit, and contingency as amounts or percentages.
4. Project Information: enter project name, client, location, and notes for the report header.

5. Select Estimate Project Cost. Read Material Cost per cubic metre, Total Material Cost, Total Project Cost, and Cost per Bag of Concrete, with the line by line breakdown table beneath.
6. Select Export Cost Report (CSV) or Export Cost Report (PDF) to save the priced bill.

7 History Tab

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Weather ⚙️

Mix Design		Material Quantification		Cost Estimation		History	
FILTER: All Types		Search by name or tag...				Search 219 records	
	ID	NAME	TYPE	DATE	KEY RESULT		
1	394	Quantification - 10.00 m³	Quantificat...	2026-09-16	Vol=10.5 m³, C=3675.0 kg		
2	393	Quantification - 10.00 m³	Quantificat...	2026-09-16	Vol=10.5 m³, C=3675.0 kg		
3	392	Cost Estimate	Cost ...	2026-09-10	Total: 88902.1875		
4	391	Cost Estimate	Cost ...	2026-09-10	Total: 87773.4375		
5	390	Cost Estimate	Cost ...	2026-09-10	Total: 1089.375		
6	389	PSD ASTM C33 Coarse — 5000.0 g	PSD Analysis	2026-09-10	Total=5000.0 g		
7	388	PSD ASTM C33 Coarse — 7900.0 g	PSD Analysis	2026-09-10	Total=7900.0 g		
8	387	PSD ASTM C33 Coarse — 5000.0 g	PSD Analysis	2026-09-10	Total=5000.0 g		
9	386	Quantification - 10.00 m³	Quantificat...	2026-09-10	Vol=10.5 m³, C=3675.0 kg		
10	385	Quantification - 10.00 m³	Quantificat...	2026-09-10	Vol=10.5 m³, C=3675.0 kg		
11	384	Mix DOE - 32.0 MPa	Mix Design	2026-09-10	W/C=0.6, C=325.0 kg [doe]		
12	383	PSD IS 383 Fine — 1000.0 g	PSD Analysis	2026-09-10	Total=1000.0 g		
13	382	Quantification - 10.00 m³	Quantificat...	2026-09-10	Vol=10.5 m³, C=3675.0 kg		
14	381	Quantification - 10.00 m³	Quantificat...	2026-09-07	Vol=10.5 m³, C=3675.0 kg		
15	380	Mix DOE - 46.0 MPa	Mix Design	2026-09-05	W/C=0.47, C=340.0 kg [doe]		
16	370	Mix DOE - 46.0 MPa	Mix Design	2026-09-05	W/C=0.47, C=340.0 kg [doe]		

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Next →

☐ Select All
 Load Selected
Delete Selected
Export Selected

Import JSON
Export All
Clear All History
Refresh

Quantified — Gross: 10,500 m³ | Cement: 3,675.0 kg (74 bags) | Water: 1,837.5 kg

Figure 14. History tab with type filter, search, records table, pagination, and batch actions.

Open the History tab to audit and reuse past work. Every Mix Design, PSD Analysis, Quantification, Cost Estimate, and DOE or ACI trial mix is stored in the local SQLite database with its inputs and key result.

- Filter: restrict the table by record type with the drop down, then type a name or tag and select Search.
- Records table: read ID, name, type, date, and key result columns. Page with Prev and Next; 50 records show per page.
- Select All and row checkboxes: tick records, then Load Selected to reopen one record back in its tab, Delete Selected to remove them, or Export Selected to save them.
- Import JSON and Export All: move records between machines or archive the full log.
- Clear All History: remove every record after confirmation. Refresh: reload after external changes.
- Double select any row to open the detail dialog with full inputs, results, and parent links for trial mixes chained to their design record.

8 Reports and Data Exchange

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Format	Where It Is Created	What It Contains
PDF report	Mix Design Export PDF, Cost Export Cost Report (PDF)	Formatted inputs, step by step calculations, quantity and cost tables, and charts.
CSV	Every tab Export CSV	Machine readable quantities, steps, sieve tables, or cost lines for spreadsheets and site logs.
JSON	History Export and Import	Full record payloads for backup, transfer, and audit.
Text report	Quantification Export Text Report	Plain text bill of materials for printing and messaging.
Image	PSD Export Image	Gradation curve chart for reports and presentations.

Always preview a PDF before saving. File dialogs default to descriptive names such as `mix_design_report.pdf`; rename per project before saving.

9 Command Line Workflow

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Run `python main_cli.py` for an interactive terminal workflow that mirrors the GUI: design a mix to ACI 211.1 or IS 10262, answer the handoff prompt, then quantify materials with editable overrides. The CLI is suited to remote servers and batch checks, while the GUI remains the primary route for charts, PSD curves, and formatted PDF reports.

10 Design Standards Referenced

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Calculations cite their clauses on screen and in reports. The three proportioning codes are ACI PRC-211.1-22 for normal and high density concrete, IS 10262:2019 with IS 456 exposure limits for Indian practice, and BRE 331:1997 DOE method for British practice. Aggregate grading follows IS 383:2016 Table 7 and Table 9 or ASTM C33/C33M Tables 1 and 2. Target strength rounds up to the whole megapascal as a conservative application policy.

Task	Reference
Target strength	IS 10262 Clause 7.1 Tables 1 and 2; BRE Section 5.1 Figure 3
Water cement ratio	ACI Sections 5 and 9; IS Table 8 Figure 1; BRE Figure 4 Section 9.3
Water content	ACI Table 5.3.3; IS Tables 7 and 12; BRE Tables 3 and 9
Aggregate proportions	ACI Tables 5.3.3 to 5.3.6; IS Tables 10 and 13; BRE Figure 6
Admixtures and SCMs	IS Annex G and Annex E; ACI Section 6.3
Moisture correction	ACI Section 9.3.8; IS Annex D-9.1
Trial batches	ACI Section 9.3.9 Appendix A.5; IS Sections 9 Annexes D E F; BRE Sections 5.6 and 7

11 Troubleshooting

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Symptom	What To Do
Calculate stays disabled	Complete all required fields; check that exposure grade and water cement ratio are consistent.
No PSD fineness modulus	The current standard and aggregate do not consume fineness modulus, or a standard sieve is missing; check the derivation note.
Transfer button greyed out	Run a successful calculation first; Send buttons activate only with a valid result.
Units look wrong	Open the gear Settings dialog and confirm Metric or Imperial before entering values.
History record will not load	Confirm the record type matches the tab; trial mixes chain to a parent design record.

PDF export fails	Choose a writable folder and a simple file name, then retry; confirm ReportLab is installed.
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CivilQntify version 1.0.10. For defects or feature requests, attach the exported CSV or JSON record and the PDF report so the calculation can be reproduced.